

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Pseudocode Writing Task (Algorithm Design)

Course Code:	DSA0307	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 2 – Pseudocode Writing Task (Algorithm Design)
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	B.E-CSE-AI	Date:	

Code of Conduct

I Shaik Nuha (192472242) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sk.Nuha

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Reference Resolution, Discourse	Entity Linking Algorithm	Q1
Text Coherence, Discourse Structure	Coherence Analysis Algorithm	Q2
Dialog Acts, Conversational Agents	Intent Recognition Algorithm	Q3
Language Generation, Surface Realization	NLG Pipeline Design	Q4
Machine Translation, Interlingua, Statistical Approaches	Translation Workflow Algorithm	Q5

Annexure A – Pseudocode Writing Task (Algorithm Design)

1. Design an algorithm to perform **Reference Resolution** by identifying pronouns and linking them to the correct entities in a discourse. Apply your algorithm to the text:

"Ravi met Arun at the library. He borrowed a book and later returned it."

Generate the resolved discourse by replacing pronouns with their corresponding entities. Further, validate the correctness of the resolution process by explaining how discourse context is used to identify the antecedents.

2. Develop an algorithm to analyze **Text Coherence** and **Discourse Structure** by identifying logical relationships between sentences such as cause-effect, sequence, and elaboration. Apply your algorithm to the following discourse:

"The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online."

Identify the discourse relations among the sentences and construct a discourse structure representation. Further, justify how these relations contribute to maintaining coherence in the text.

3. Design an algorithm for a **Conversational Agent** that identifies **Dialogue Acts** from user interactions. Apply your algorithm to the conversation:

User: "Can you book a train ticket for me?"

Agent: "Sure, where would you like to travel?"

User: "I want to go to Chennai."

Agent: "Your ticket has been booked."

Classify each utterance as Request, Question, Inform, Confirmation, or Action. Generate the dialogue-act sequence and evaluate how the identified acts help the conversational agent interpret user intentions.

4. Design an algorithm for **Language Generation** that converts a semantic representation into a grammatically correct sentence through **Surface Realization**. Consider the semantic input:

Action: Buy
Agent: Student
Object: Book
Tense: Past

Generate the final natural language sentence and explain how the algorithm performs lexical selection, sentence structuring, and surface realization. Further, validate the grammatical correctness of the generated output.

5. Design an algorithm that combines **Interlingua-based Machine Translation** and **Statistical Translation Approaches**. Apply your algorithm to translate the sentence:

"The boy is playing football."

Perform the following steps:

1. Analyze the source sentence.
2. Create an Interlingua representation.
3. Generate candidate target-language translations.
4. Apply statistical scoring to select the most appropriate translation.
5. Produce the final translated sentence.

Finally, evaluate how the Interlingua representation and statistical methods improve translation quality and reduce ambiguity.

Rubrics for Calculation

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

Q. No. / Criteria Level	Problem Understanding	Algorithm Design	Use of Control Structures	Pseudocode Structure	Completeness	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

1. Reference Resolution Algorithm

Problem Statement

Design an algorithm that identifies pronouns in a discourse and links them to their correct antecedent entities.

Given:

“Ravi met Arun at the library. He borrowed a book and later returned it.”

We need to identify what “He” and “it” refer to and then replace them with the appropriate entities.

1.1 Introduction

Reference Resolution is the NLP task of identifying the entity to which a referring expression, especially a pronoun, refers.

For example:

Ravi met Arun. He borrowed a book.

The system must determine whether “He” refers to Ravi or Arun.

A reference-resolution algorithm generally uses:

- Pronoun identification
- Gender agreement
- Number agreement
- Recency
- Grammatical compatibility
- Semantic compatibility
- Discourse context

1.2 Proposed Algorithm

Input

A discourse containing sentences and referring expressions.

Output

Resolved references and a rewritten discourse with explicit entities.

Algorithm

Algorithm Reference Resolution (Text)

1. Input the discourse.
2. Split the discourse into sentences.
3. Perform tokenization and part-of-speech tagging.
4. Identify all noun phrases/entities.
5. Identify pronouns.
6. For each pronoun:
 - a. Generate possible antecedents from previous context.
 - b. Check gender agreement.

- c. Check number agreement.
 - d. Check grammatical compatibility.
 - e. Check semantic compatibility.
 - f. Apply recency preference.
 - g. Check discourse coherence.
7. Assign the highest-scoring compatible antecedent.
 8. Add the pronoun and antecedent to the same entity chain.
 9. Replace the pronoun with its resolved entity.
 10. Return the resolved discourse.

1.3 Applying the Algorithm

Sentence 1

Ravi met Arun at the library.

The entities introduced are:

$$E = \{\text{Ravi, Arun, Library}\}$$

Ravi and Arun are both potential antecedents for a later masculine pronoun.

Sentence 2

He borrowed a book and later returned it.

There are two pronouns:

1. He
2. It →

1.4 Resolving “He”

Possible antecedents:

- Ravi
- Arun

Both are:

- Singular
- Male
- Human

Therefore, gender and number constraints cannot uniquely resolve “He.”

We need discourse-level information.

Recency

Arun occurs immediately before the pronoun:

Ravi met Arun at the library. He...

Therefore, recency gives Arun a stronger preference.

Semantic Compatibility

The action:

“borrowed a book” is compatible with both Ravi and Arun.

Therefore, semantic compatibility does not eliminate either candidate.

Discourse Coherence

Since Arun is the most recent compatible human male entity, the system selects Arun.

He →Arun

1.5 Resolving “it”

The sentence says:

He borrowed a book and later returned it.

Possible antecedents include:

- Ravi
- Arun
- Book

The pronoun “it” is singular and normally refers to an inanimate object in this context.

The phrase:

“returned it”

is strongly compatible with:

“a book”

Therefore:

it →book

1.6 Entity Chains

The resulting entity chains are:

Entity Chain 1

Arun →He

Entity Chain 2

Book →it

Ravi does not have a later referring expression.

1.8 Validation

Pronoun	Candidate	Agreement	Recency	Semantic Compatibility	Result
He	Ravi	✓	Lower	✓	Possible
He	Arun	✓	Higher	✓	Selected
it	Ravi	X/Weak	—	X	Rejected
it	Arun	X/Weak	—	X	Rejected
it	Book	✓	High	✓✓	Selected

Final Answer

He=Arun

it=Book

2. Text Coherence and Discourse Structure

Given Discourse

“The roads were flooded after heavy rainfall. Therefore, schools were closed for the day. Students attended classes online.”

2.1 Introduction

Text coherence refers to the logical and meaningful connection between different parts of a text.

A coherent discourse should have relationships such as:

- Cause → Effect
- Sequence
- Elaboration
- Contrast
- Condition
- Explanation

In this example, the sentences form a clear causal and sequential chain.

2.2 Proposed Algorithm

Algorithm Discourse Coherence (Text)

1. Split the text into sentences.
2. Identify important entities and events.
3. Extract predicates and actions from each sentence.
4. Identify discourse markers such as:
therefore, because, then, however, additionally, etc.
5. Determine semantic relationships between adjacent sentences.
6. Classify relations as:
Cause-Effect
Sequence
Elaboration
Contrast
Condition
7. Construct a discourse graph/tree.
8. Validate whether each sentence logically connects to the previous sentence.
9. Output the discourse structure.

2.3 Sentence Analysis

Sentence 1

The roads were flooded after heavy rainfall.

Important events:

Heavy Rainfall → Flooding

The heavy rainfall causes the roads to flood.

Sentence 2

Therefore, schools were closed for the day.

Therefore is a strong discourse marker indicating a Cause-Effect relationship.

The reasoning is:

Flooded Roads → Difficult/Unsafe Travel → Schools \ Closed

Sentence 3

Students attended classes online.

This sentence describes what happened after schools were closed.

Therefore, it represents a Sequence/Result relationship.

S2 → S3

There is also a practical consequence:

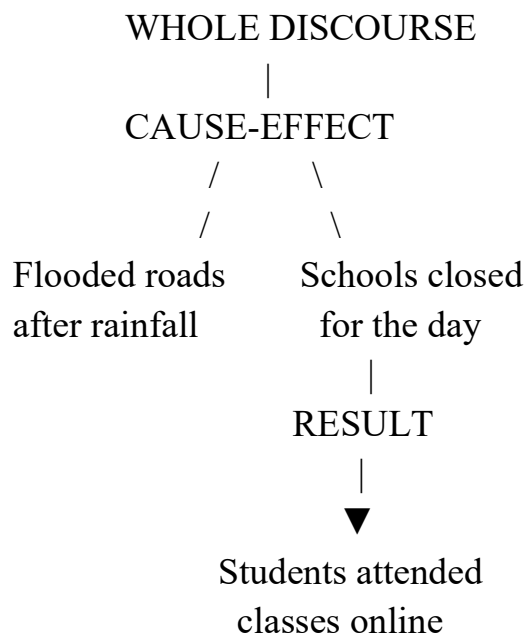
Schools Closed → Students Attend Online

2.4 Discourse Relations

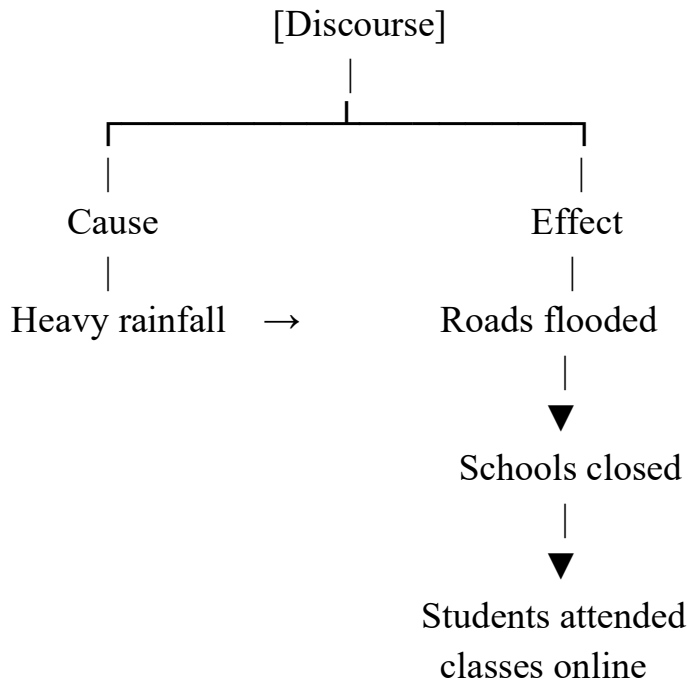
Sentence Pair	Relation	Explanation
S1 → S2	Cause-Effect	Flooded roads caused school closure
S2 → S3	Result/Sequence	School closure led students to attend online
S1 → S3	Indirect Cause-Effect	Rainfall indirectly caused online classes

2.5 Discourse Structure Representation

A simple tree:



A more detailed representation:



Conclusion:

The discourse contains:

Cause-Effect + Sequence/Result

These relations ensure that the reader understands why schools were closed and what happened afterward.

3. Dialogue Act Identification for a Conversational Agent

Given Conversation

User: “Can you book a train ticket for me?”

Agent: “Sure, where would you like to travel?”

User: “I want to go to Chennai.”

Agent: “Your ticket has been booked.”

3.1 Introduction

A Dialogue Act represents the communicative intention of an utterance.

Common dialogue acts include:

- Request
- Question
- Inform
- Confirmation
- Action

Identifying dialogue acts helps a conversational agent understand what the user wants and determine the next appropriate response.

3.2 Proposed Algorithm

Algorithm Dialogue Act Detection (Utterance)

1. Receive the user/agent utterance.
2. Perform tokenization and linguistic analysis.
3. Identify keywords and sentence structure.
4. Detect communicative intention.
5. Classify the utterance as:
 - Request
 - Question
 - Inform
 - Confirmation
 - Action
6. Store the detected act in the dialogue state.
7. Use the dialogue state to determine the next response.
8. Repeat until the task is completed.

3.3 Utterance 1

User:

“Can you book a train ticket for me?”

The user is asking the system to perform a task.

Therefore:

Request

The user's underlying intention is:

BookTrainTicket

3.4 Utterance 2

Agent:

“Sure, where would you like to travel?”

The agent asks for missing information.

Therefore:

Question

The agent needs the destination before completing the booking.

3.5 Utterance 3

User:

“I want to go to Chennai.”

The user provides the requested information.

Therefore:

Inform

The information supplied is:

Destination=Chennai

3.6 Utterance 4

Agent:

“Your ticket has been booked.”

The agent indicates that the requested action has been completed.

This can be classified as:

Confirmation

It also represents completion of the booking action.

Therefore, depending on the classification scheme:

Confirmation + Action

3.7 Dialogue-Act Sequence

The main sequence is:

Request → Question → Inform → Confirmation

If the actual system action is explicitly represented:

Request → Question → Inform → Action → Confirmation

3.8 Dialogue State

The agent can maintain a state representation:

Task = BookTrainTicket

Destination = Chennai

Status = Booked

Initially:

Task = BookTrainTicket

Destination = Unknown

Status = Pending

After the user's information:

Task = BookTrainTicket

Destination = Chennai

Status = Pending

After booking:

Task = BookTrainTicket

Destination = Chennai

Status = Booked

3.9 Why Dialogue Acts Are Important

Dialogue acts allow the agent to understand user intention.

For example:

Request

Tells the agent:

“The user wants me to perform an action.”

Question

Tells the system:

“Information is required.”

Inform

Tells the system:

“The user has provided the required information.”

Confirmation

Tells the user:

“The requested operation has been completed.”

Therefore:

Dialogue Acts → Intent \ Understanding → Dialogue \ Management → Correct \ Response

4. Language Generation and Surface Realization

Semantic Input

Action: Buy

Agent: Student

Object: Book

Tense: Past

The objective is to convert this semantic representation into a grammatically correct sentence.

4.1 Introduction

Natural Language Generation (NLG) converts structured meaning into natural language.

The process generally includes:

1. Semantic interpretation
2. Lexical selection
3. Sentence planning
4. Syntactic structuring
5. Morphological realization
6. Surface realization

4.2 Proposed Algorithm

Algorithm SurfaceRealization(SemanticInput)

1. Read semantic roles.
2. Identify the main action.
3. Select lexical items.
4. Assign semantic roles:
 - Agent → Subject
 - Action → Verb
 - Object → Direct Object

5. Determine tense.
6. Apply grammatical rules.
7. Perform morphological realization.
8. Arrange words in target language order.
9. Check grammatical agreement.
10. Generate final sentence.

4.3 Step 1 — Identify Semantic Roles

Input:

Agent=Student
Action=Buy
Object=Book
Tense=Past

Therefore, the basic structure is:

Subject + Verb + Object

4.4 Step 2 — Lexical Selection

The semantic concepts are mapped to words:

Semantic Concept	Lexical Item
Agent	student
Action	bought
Object	book

The verb buy must be converted into its past-tense form.

Buy → Bought

This is an example of morphological realization.

4.5 Step 3 — Sentence Structure

English commonly follows:

SVO

where:

- S = Subject
- V = Verb
- O = Object

Therefore:

Student + Bought + Book

4.6 Final Sentence

The student bought a book.

4.7 Surface Realization Pipeline

Semantic Representation



Role Assignment



Lexical Selection



Syntactic Structure



Tense/Morphology



Determiner Selection



Surface Realization



"The student bought a book."

5. Hybrid Interlingua and Statistical Machine Translation

Given Sentence

"The boy is playing football."

The objective is to combine:

- Interlingua-based translation
- Statistical translation

5.1 Introduction

Machine Translation converts text from a source language into a target language.

Two important approaches are:

Interlingua

The source sentence is converted into a language-independent semantic representation.

Statistical Translation

Multiple possible target translations are generated and assigned probabilities based on learned language and translation models.

A hybrid system combines the advantages of both.

5.2 Proposed Algorithm

Algorithm HybridMachineTranslation(SourceSentence)

1. Analyze source sentence.
2. Perform tokenization and POS tagging.
3. Identify grammatical structure.
4. Extract semantic roles.

5. Convert sentence into Interlingua representation.
6. Generate candidate target-language translations.
7. Calculate statistical translation scores.
8. Calculate language-model scores.
9. Combine semantic and statistical scores.
10. Select the highest-scoring valid translation.
11. Perform grammatical checking.
12. Output final translation.

5.3 Step 1 — Source Sentence Analysis

Sentence:

The boy is playing football.

Tokens

The, boy, is, playing, football

Parts of Speech

Word	POS
The	Determiner
boy	Noun
is	Auxiliary verb
playing	Verb
football	Noun

5.4 Syntactic Structure

The sentence follows:

Subject + Verb + Object

Subject:

The boy

Verb phrase:

is playing

Object:

football

The tense aspect is:

Present\ Progressive

5.5 Semantic Roles

We can represent the meaning as:

Agent=Boy

Action=Play

Object=Football

Tense=Present

Aspect=Progressive

5.6 Step 2 — Interlingua Representation

An interlingua representation can be:

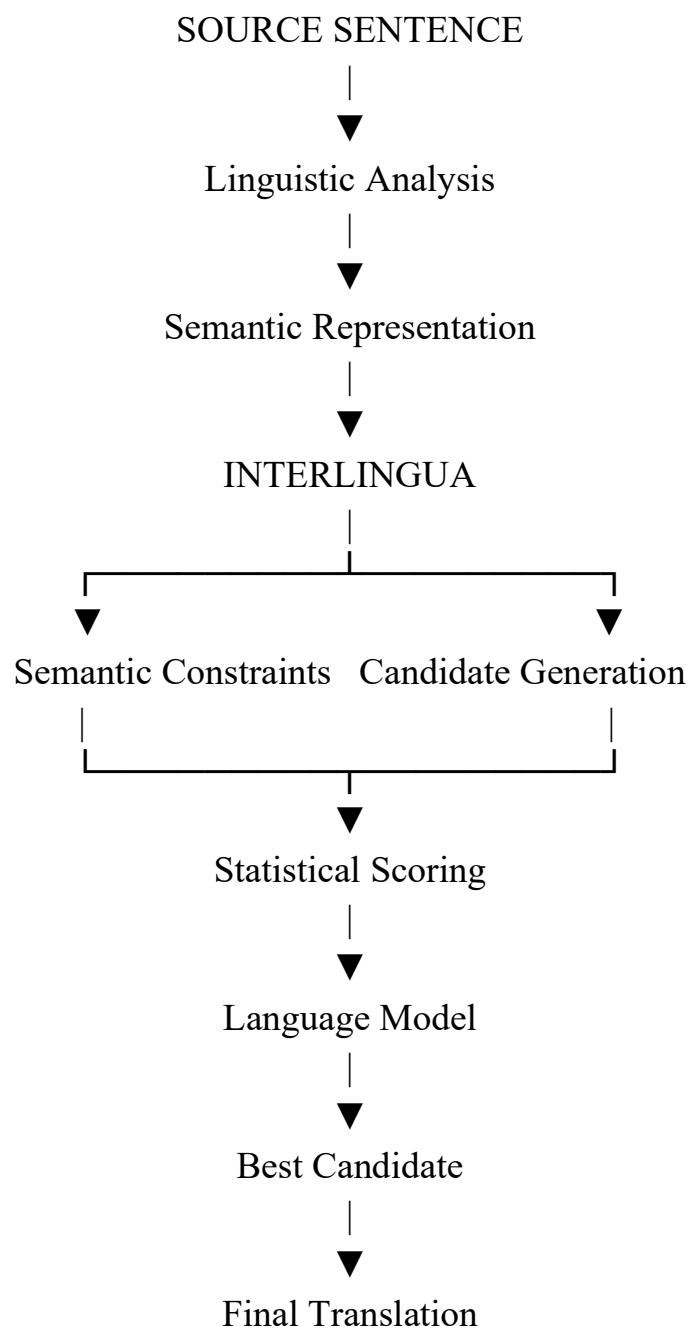
```
PLAY (  
  AGENT = BOY,  
  OBJECT = FOOTBALL,  
  TENSE = PRESENT,  
  ASPECT = PROGRESSIVE  
)
```

Or using predicate logic:

Playing (Boy, Football, Present Progressive)

The important point is that this representation is independent of the target language.

5.7 Hybrid Architecture



5.8 Why Statistical Translation Helps

Statistical methods are useful because there may be several grammatically possible target-language expressions.

The statistical component can consider:

- Word translation probability
- Phrase probability
- Word order
- Target-language fluency
- Context
- Common usage

Therefore, the system can choose the most natural candidate.

5.9 Evaluation of the Hybrid Approach

Advantage 1 — Reduced Ambiguity

Interlingua explicitly represents the meaning.

For example:

Action=Play

Agent=Boy

Object=Football

This reduces the chance that the target-language system changes the intended roles.

Advantage 2 — Better Semantic Preservation

The semantic structure is maintained across translation.

Boy → Agent

Football → Object

Play → Action

Advantage 3 — Language Independence

The interlingua representation does not depend heavily on English word order.

Therefore, it can be mapped into languages with different grammatical structures.